

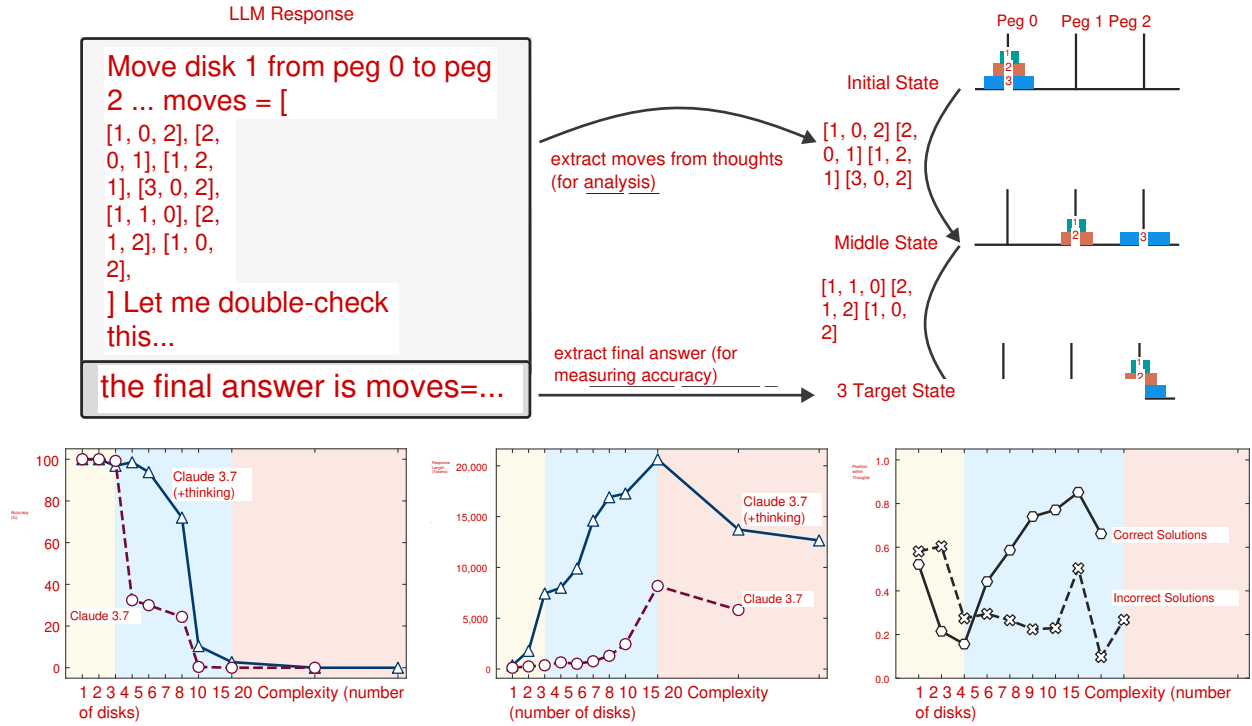


Figure 1: Top: Our setup enables verification of both final answers and intermediate reasoning traces, allowing detailed analysis of model thinking behavior. Bottom left & middle: At low complexity, non-thinking models are more accurate and token-efficient. As complexity increases, reasoning models outperform but require more tokens—until both collapse beyond a critical threshold, with shorter traces. Bottom right: For correctly solved cases, Claude 3.7 Thinking tends to find answers early at low complexity and later at higher complexity. In failed cases, it often fixates on an early wrong answer, wasting the remaining token budget. Both cases reveal inefficiencies in the reasoning process.

emergence suggests a potential paradigm shift in how LLM systems approach complex reasoning and problem-solving tasks, with some researchers proposing them as significant steps toward more general artificial intelligence capabilities. Despite these claims and performance advancements, the fundamental benefits and limitations of LRMs remain insufficiently understood. Critical questions still persist: Are these models capable of generalizable reasoning, or are they leveraging different forms of pattern matching [6]? How does their performance scale with increasing problem complexity? How do they compare to their non-thinking standard LLM counterparts when provided with the same inference token compute? Most importantly, what are the inherent limitations of current reasoning approaches, and what improvements might be necessary to advance toward more robust reasoning capabilities? We believe the lack of systematic analyses investigating these questions is due to limitations in current evaluation paradigms. Existing evaluations predominantly focus on established mathematical and coding benchmarks, which, while valuable, often suffer from data contamination issues and do not allow for controlled experimental conditions across different settings and complexities. Moreover, these evaluations do not provide insights into the structure and quality of reasoning traces. To understand the reasoning behavior of these models more rigorously, we need environments that enable controlled experimentation. In this study, we probe the reasoning mechanisms of frontier LRMs through the lens of problem